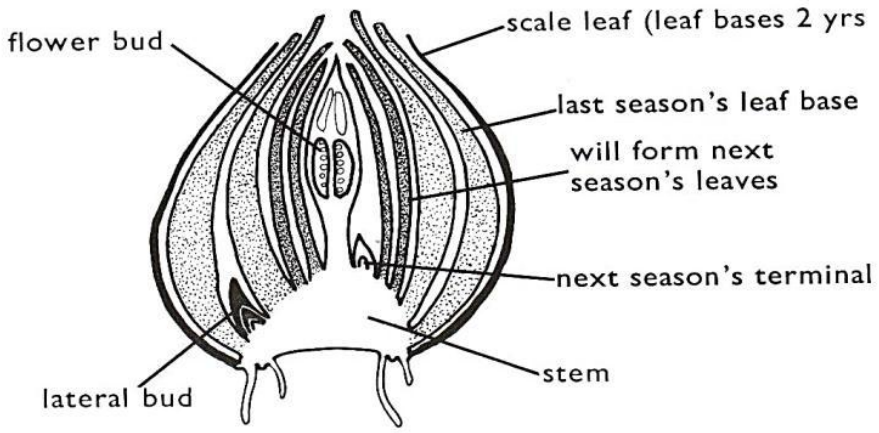
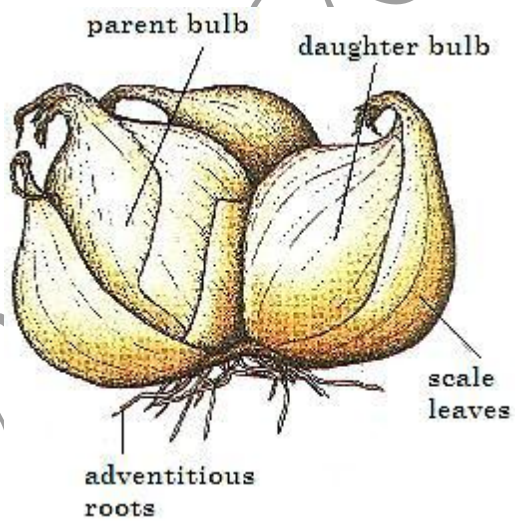
BULB

It is a shortened, compressed underground stem surrounded by fleshy leaves that develop a central bud located at the tip of the stem. It includes onions, tulips, lilies etc. The fleshy leaves are the storage organs that store food in form of starch.





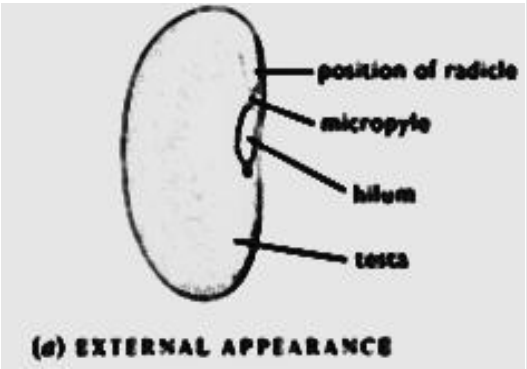
FRUITS AND SEEDS

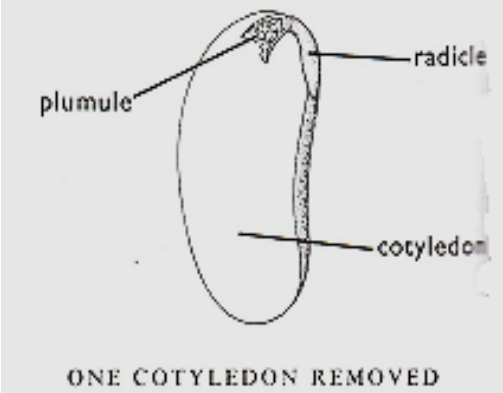
A fruit is a fertilized ovary of a flower. It is generally formed from the ovary and the ovules, e.g. mango, tomato, maize seed. The ovary forms the fruit wall called the pericarp. The pericarp is made up of three parts namely the epicarp, fleshy mesocarp and the endocarp. Inside the endocarp is the seed.

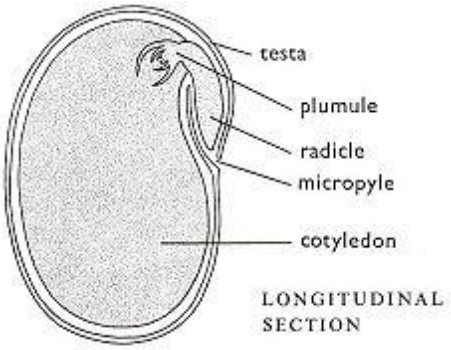
A seed develops from an ovule after fertilization. It consists of a tough coat or testa enclosing the embryo which is made up of a plumule, a radical and one or two cotyledons. In favourable conditions the seed germinate and grows into an independent plant bearing flowers and seeds.

Parts Of A Seed or Fruit and its Functions

a. Bean Seed







TUBERS

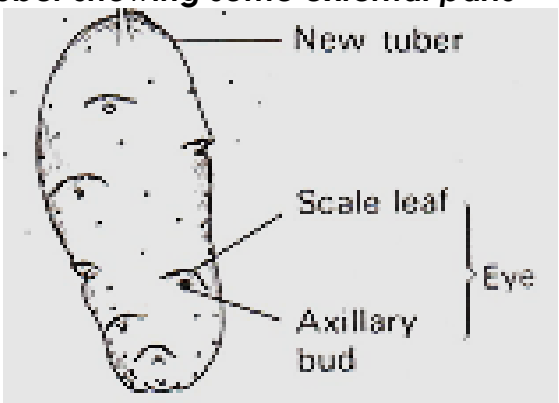
A tuber is an enlarged portion of an underground stem like in potato tuber or an enlarged portion of an underground root like cassava.

It stores food in form of starch because this is in the insoluble state where it cannot easily dissolve in water.

The tuber like any other stem has nodes that produce buds. The eyes of a potato are actually the nodes on the stem.

Each eye contains a cluster of buds.

Diagram of a potato tuber showing some external parts



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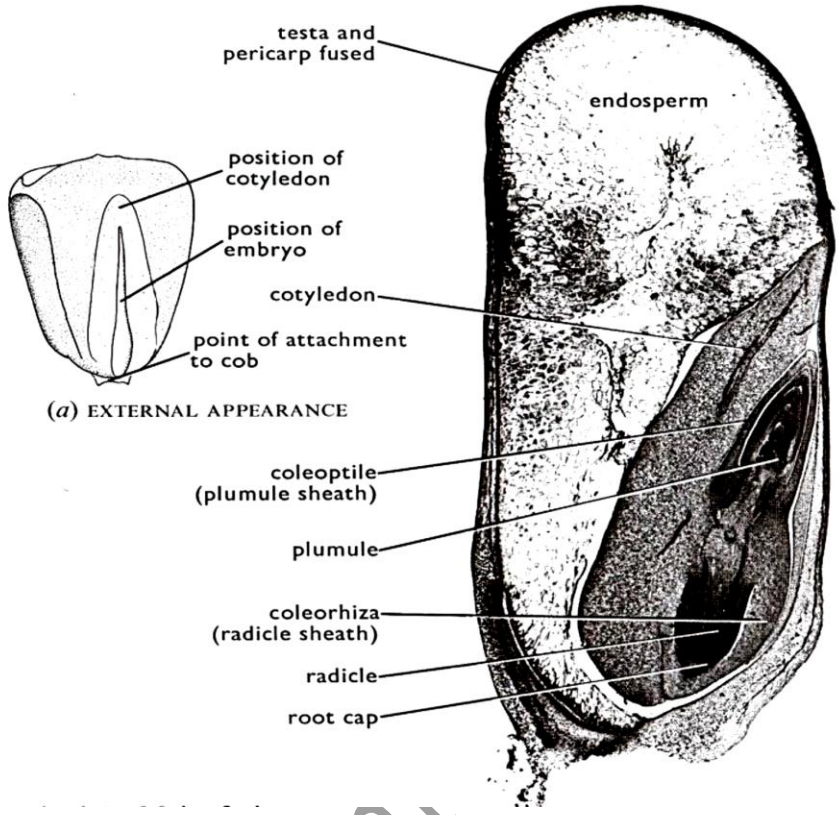
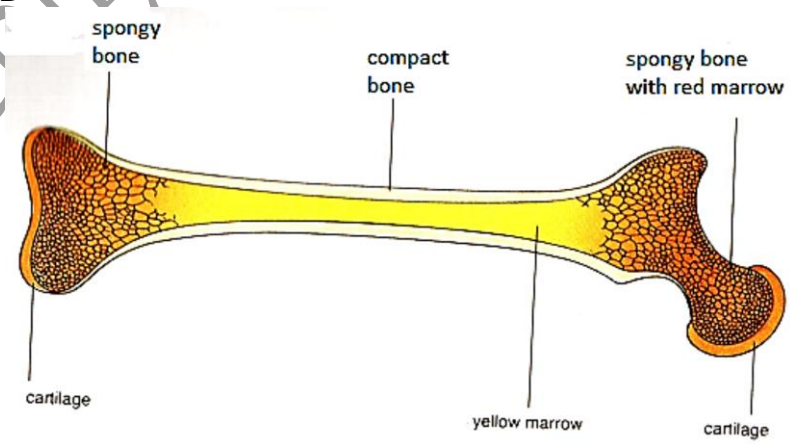
MAIZE SEED



STRUCTURE AND COMPOSITION OF A BONE

Formation of bones in the body requires the following;

- Calcium
- Amino acids
- Phosphate
- Hormones
- Vitamin D



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CARTILAGE
This is a hard but flexible supporting tissue found at the ends of bones. Its function is to prevent the bones from rubbing against each other. It also enables the body to withstand shocks and jar rings that accompany movements.
COMPACT BONE
This is the hardest part of a bone that provides strength and stores calcium.
BONE MARROW
It is a soft tissue with a good supply of blood. Its functions includes manufacturing red blood cells, white blood cells and platelets.

SPONGY BONE
It has spaces in it which stops the bone from being too heavy.

NB: Periosteum is a structure where tendons are attached.
A bone is made up of the compact bone which is the hardest, the spongy bone with spaces and the bone marrow which is very soft and has a good supply of blood.
A bone is composed of organic and inorganic substances. The organic component is made of a protein collagen which makes it flexible and rubbery. The inorganic component is made up of calcium phosphate and calcium carbonate.
Strong bones have a hole and are curved. This enables them to support the weight of the body.

VITAMIN C TEST

- *Place 2cm3 of the dye of dichlorophenolindophenol (DCPIP)in a test tube*
- *Add1% vitamin C solution a drop at a time from a graduated pipette*
- *The dye decolorizes (becomes colourless)*
- *Presence of vitamin C*

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